

Problem Set 2 – Solution

Microeconomic Theory 3 – ECON 709, Fall 2026

This document provides a solution sketch for each answer. Sometimes, the suggested solution is of course not the only "right" answer, but you might have come up with different, equally correct answers.

1. (Rubinstein, Lecture 2, Problem 2. Solution from Rubinstein's solution manual.) Give an example of preferences over a countable set in which the preferences cannot be represented by a utility function that returns only integers as values.

Solution:

Of course, if the preferences are not rational, this is trivially impossible.

Suppose the preferences are rational. Consider $X = \mathbb{N}$, which is countable. Define the preference relation such that you prefer all odd numbers strictly to all even numbers, and among the odd numbers as well as the even numbers you strictly prefer a lower one to a higher one, that is

$$1 \succ 3 \succ 5 \succ \dots \succ 2 \succ 4 \succ 6 \dots$$

By contradiction, assume that there exists a utility function $u : X \rightarrow \mathbb{Z}$ that represents $\succsim$. Then $u(1) = N$ and $u(2) = n$ for some $n, N \in \mathbb{Z}$. But there are infinite odd numbers, implying that u maps to an infinite number of integers between n and N , a contradiction, because there are finitely many integers between n and N .

2. (Rubinstein, Lecture 2, Problem 5. The solution is taken from Rubinstein's solution manual.) Let X be a set and let $(\succsim, \succ)$ be a pair where $\succsim$ is a rational preference relation; let $\succ$ be induced from $\succsim$, and let $\succ\succ$ be a transitive subrelation of $\succ$ (by subrelation, we mean that $x \succ\succ y$ implies $x \succ y$.) The meaning of $x \succ\succ y$ is that x is much

preferred over y . Assume that the pair satisfies the following extended transitivity condition: If $x \succ y$ and $y \succsim z$, or if $x \succsim y$ and $y \succ z$, then $x \succ z$.

We say that a pair $(\succsim, \succ)$ is represented by a function u if:

$$u(x) = u(y) \text{ iff } x \sim y,$$

$$u(x) - u(y) > 0 \text{ iff } x \succ y, \text{ and}$$

$$u(x) - u(y) > 1 \text{ iff } x \succ y.$$

Show that if X is finite every extended preference $(\succsim, \succ)$ can be represented by a function u .

Solution:

Denote by $A \succ B$ if $a \succ b$ for all $a \in A$ and all $b \in B$. Let $X_1, X_2, \dots, X_K$ be the indifference sets given $\succsim$ such that $X_K \succ X_{K-1} \succ \dots \succ X_1$.

Define first $u(X_1) = 0$.

Let us define $u(X_k)$ for $k > 1$.

(1) If $X_k \succ X_{k-1}$, then $u(X_k) = u(X_{k-1}) + 1$.

(2) If there is no $l < k$ (not even $l = 1$) such that $X_k \succ X_l$, then $u(X_k) \in (u(X_{k-1}), 1)$.

(3) Otherwise, there exists a maximal $m(k)$ such that $X_k \succ X_{m(k)}$. In this case, define $u(X_k)$ such that $u(X_k) > u(X_{k-1})$ and $1 + u(X_{m(k)+1}) > u(X_k) > u(X_{m(k)}) + 1$.

Clearly, $x \sim y$ iff $u(x) = u(y)$.

Also, if $x \succ y$ then $u(x) > u(y)$, since we picked $u(X_k)$ as an increasing sequence.

Finally, if $x \succ y$, $x \in X_k$ and $y \in X_m$ then $m(k) \geq m$ and $u(x) > u(X_{m(k)}) + 1 \geq u(y) + 1$.